MULTI-STEP AI WORKFLOW

Assessment Submission

Scenario: AI Study Planner — From Student Requirements to a Refined Personalized Plan

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Assessment Format	Chained multi-prompt AI workflow

Submission Note

This document presents a complete three-stage chained workflow. Each stage explicitly depends on the output of the previous stage, and the final stage includes a self-critique gate before producing the refined answer.

Font: Times New Roman-compatible serif formatting throughout the document.

1. Assessment Understanding & Workflow Design

The provided assessment rules require a multi-stage task with at least three distinct prompts in the same chat session. The prompts must be chained, meaning each later prompt must use, critique, or transform the previous output. At least one step must act as a critique/refinement gate, and the submission must document the prompt transcript and intermediate responses. fileciteturn0file1L5-L12

The reference example demonstrates a clear progression: planning first, drafting second, and evaluation/refinement third. fileciteturn0file0L6-L18

Selected Scenario

AI Study Planner: Convert a student's academic constraints into a practical weekly study plan, generate a detailed schedule from that plan, then audit and improve the schedule against explicit criteria.

Stage	Purpose	Input	Output
1 — Planning	Extract priorities, constraints and planning rules.	Student requirements	Structured study-plan blueprint
2 — Generation	Turn the blueprint into a usable weekly timetable.	Stage 1 blueprint	Detailed weekly schedule
3 — Critique & Refinement	Audit the schedule and correct weaknesses.	Stage 2 schedule + checklist	Critique + final refined schedule

Why this scenario is suitable

It is naturally multi-stage: a good timetable should not be generated blindly. The first stage identifies what matters, the second operationalizes those decisions, and the third checks whether the result is balanced, realistic and complete. This creates a visible dependency between all three prompts rather than three unrelated questions.

2. Exact Prompt Chain Transcript

The following are the exact prompts used for the three-stage workflow. The wording intentionally makes the dependency explicit.

Prompt 1 — Extraction & Planning

> Act as an expert academic planning assistant. I am a CSE-AIML student preparing for multiple subjects and technical interviews. I have 7 days available. My priorities are: Java programming, SQL/DBMS, DSA, AI/ML fundamentals, and project work. I can study about 4 focused hours on weekdays and 6 hours on the weekend. I prefer shorter sessions with breaks and I want daily revision rather than leaving revision to the final day.
>
> Task: Extract the important requirements, identify priorities, create a structured weekly planning blueprint, and assign a reasonable time allocation to each area. Do not create the final timetable yet. Your output must be organized as a table followed by 4–6 planning rules that can be directly used in the next prompt.

Prompt 2 — Generation / Drafting

> Using only the planning blueprint and planning rules produced in Step 1, create the first draft of a 7-day study timetable. Preserve the priorities and time allocations from Step 1.
>
> Constraints: 4 focused hours Monday–Friday and 6 focused hours Saturday–Sunday; include breaks between sessions; include daily revision; give every listed subject a place; keep sessions practical rather than overly long; include a small buffer for project work. Present the result as a day-by-day table with time blocks and a short explanation of the strategy.

Prompt 3 — Critique Gate & Refinement

> Before finalizing the timetable, act as a strict academic reviewer. Review the Step 2 draft against this checklist: (1) weekday/weekend hour limits are respected, (2) all five priority areas receive meaningful time, (3) every day contains revision, (4) sessions are not excessively long, (5) project work has a realistic buffer, and (6) the sequence supports learning rather than random topic switching.
>
> First, list any failures or weak points and explain the correction. Second, output the complete corrected timetable. Do not remove useful content unless necessary; improve the existing Step 2 schedule using the checklist.

3. Intermediate AI Responses

Step 1 Output — Planning Blueprint

Area	Priority	Weekly Focus	Planning Rule
Java	High	7–8 hrs	Practice code daily; increase difficulty gradually.
DSA	High	6–7 hrs	Combine concept review with problem solving.
SQL / DBMS	High	5–6 hrs	Alternate SQL practice and DBMS concepts.
AI / ML	Medium	4–5 hrs	Use concept → example → short recall pattern.
Project	Medium	3–4 hrs	Keep a small, protected buffer instead of one large block.

Planning rules generated by Step 1: (1) Put the hardest technical work earlier in the day. (2) Use 50–60 minute focus blocks with short breaks. (3) End each day with active recall or revision. (4) Avoid stacking similar difficult topics for too long. (5) Keep project work in manageable blocks. (6) Use the weekend for deeper problem solving and consolidation.

Step 2 Output — First Draft Schedule

Day	Focus Blocks	Revision / Buffer
Mon	Java practice • DSA concepts • SQL practice • AI/ML basics	20 min Java + SQL recall
Tue	DSA problems • Java OOP • DBMS concepts • Project	20 min DSA recall
Wed	SQL queries • Java coding • AI/ML concepts • DSA problems	20 min mixed recall
Thu	DSA problems • DBMS • Java coding • Project	20 min Java/DBMS recall
Fri	Java interview practice • SQL • AI/ML • DSA	20 min weekly weak-area recall
Sat	Deep DSA • Java project coding • SQL/DBMS • AI/ML	30 min revision + project buffer
Sun	Mock coding • SQL test • AI/ML review • Project • Weekly recap	Final weekly review + next-week notes

4. Critique Gate & Final Refined Output

Step 3 Self-Critique

Checklist	Review	Action
Hour limits	The draft is close, but revision blocks need to be counted separately to avoid exceeding the focused-hour target.	Treat revision as part of the daily focus allocation and keep total focused study within the stated limit.
All priority areas	All five areas appear, but project work is less consistent than the other areas.	Add a protected project block on two weekdays and one weekend block.
Daily revision	Revision is present every day.	Keep it, but use active recall rather than rereading.
Session length	The draft does not define exact block lengths.	Use 50-minute focus + 10-minute break cycles.
Learning sequence	The sequence is generally balanced but can better connect concepts to practice.	Place concept learning immediately before related practice where possible.

Final Corrected 7-Day Study Plan

Day	50-min Focus Sessions	10-min Breaks / End Review
Monday 4 hrs	1. Java coding 2. DSA concepts + 2 problems 3. SQL query practice 4. AI/ML fundamentals	Break after each session; final 20 min = active recall
Tuesday 4 hrs	1. DSA problems 2. Java OOP 3. DBMS concepts 4. Project implementation	Short breaks; final 20 min = DSA + DBMS recall
Wednesday 4 hrs	1. SQL practice 2. Java coding 3. AI/ML concepts + examples 4. DSA problems	Short breaks; final 20 min = mixed recall
Thursday 4 hrs	1. DSA problems 2. DBMS + SQL 3. Java interview questions 4. Project implementation	Short breaks; final 20 min = Java/DBMS recall
Friday 4 hrs	1. Java mock coding 2. SQL test 3. AI/ML revision 4. DSA timed practice	Short breaks; final 20 min = identify weak areas
Saturday 6 hrs	1. Deep DSA practice 2. Java problem set 3. SQL/DBMS test 4. AI/ML application 5. Project work 6. Weak-area repair	Break after each block; final 30 min = weekly consolidation

Day	50-min Focus Sessions	10-min Breaks / End Review
Sunday 6 hrs	1. Coding mock test 2. SQL/DBMS mock test 3. AI/ML review 4. Project completion 5. DSA revision 6. Weekly recap + next-week planning	Break after each block; final 30 min = recall + plan

Refinement result: The final schedule retains the original Step 1 priorities, converts them into concrete sessions, adds protected project time, uses consistent focus cycles, and makes daily revision explicit. This demonstrates the required evaluation-and-refinement gate.

5. Documentation, Dependency Check & Submission Evidence

The assessment asks students to compile the exact prompt chain, intermediate AI responses, and direct evidence such as a chat link or screenshot. fileciteturn0file1L31-L35

Dependency Verification

Requirement	How this submission satisfies it
At least 3 chained prompts	Three prompts are included: planning → generation → critique/refinement.
Explicit dependency	Prompt 2 instructs the model to use only the Step 1 blueprint; Prompt 3 audits and improves the Step 2 draft.
Critique gate	Prompt 3 contains six explicit checklist criteria and requires critique before the corrected final schedule.
Intermediate outputs	Step 1 planning output and Step 2 draft are documented before the final result.
Originality / live execution	The workflow is designed for sequential execution in the assigned AI interface; live evidence should be captured from that interface.
Submission evidence	Prompt transcript and outputs are documented here. A screenshot or chat link from the live AI session should be attached if the evaluator strictly requires direct UI evidence.

Live Evidence Placeholder

Paste the live AI-chat screenshot here

Recommended evidence: one screenshot showing the sequential conversation with Prompt 1, its response, Prompt 2, its response, Prompt 3, and the final refined response. Do not use a fabricated screenshot; capture the actual assigned AI interface after running the three prompts.

Final submission checklist: verify student details → verify all three prompts → verify intermediate responses → add live screenshot/chat link if required → export/submit as PDF.

End of Assessment Submission